Alicia M. Chun

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Portfolio: https://aliciamchun.github.io/ (in **HTML/CSS**)

PROFESSIONAL STATEMENT

Detail-driven Computer Science undergraduate with specialized experience in hardware integration and human-computer design. Experienced in PCB design and microcontroller development using C/C++, with a track record of success in research positions at Subaru Telescope and NASA's L'SPACE Academy. Strong people skills with a passion for creating user-centered embedded systems that exhibit practical, human-focused engineering solutions.

EDUCATION

The University of Chicago — Chicago, IL

B.S. in Computer Science; Minor in Physics — June 2025

Cumulative GPA: 3.739/4.0; President's Scholar, University Scholar Award

Relevant Coursework: Engineering Printed Circuit Boards; Mobile Computing; Foundations of Computer Networks; Computer Architecture; Electronics; Engineering Interactive Devices; Theory of Algorithms, Adversarial Machine Learning, Quantitative Portfolio Management and Algorithmic Trading

WORK EXPERIENCE

Xylem — Remote

Product Engineer in Engineering Leadership Development Program; 40 hrs/week — July 2025 – Present

- Develop embedded firmware in **C** for AquaCase products, implementing NFC logic and multi-language functionality.
- Design control boards and user interfaces for the SOLO3 product line using **Altium Designer**.
- Collaborate with sales and supply chain teams to source and qualify hardware components for new product designs.

NASA L'SPACE Program, Mission Concept Academy — Remote

Participant - Team Role: Project Manager; 20 hrs/week — Sept. 2024 – Jan. 2025

- Virtual NASA workforce preparation academy that teaches space mission concept formulation.
- Written and presented a **183-page Preliminary Design Review** proposal using Technical Writing for the NASA mission life cycle of a robotic space mission managing mission requirements.
- Managed a team of 15 and organized the team's operations and procedures, such as: scheduling weekly meetings, leading task distribution and progress checks, maintaining consistency across deliverables.

The University of Chicago, Department of Physics — Chicago, IL

Learning Assistant; 10 hrs/week — Sept. 2023 – May 2025

- Facilitate small labs with ~20 students for the introductory physics and engineering courses.
- Serviced 3D printers, metalworking equipment, and helped students with soldering and circuit debugging.
- Specific courses: Mechanics (Autumn 2023); Electricity and Magnetism (Winter 2024); Waves, Optics, and Heat (Spring 2024); Creative Machines and Innovative Instrumentation (Winter 2025, Spring 2025)

Human Computer Integration Lab, Computer Science Department, UChicago — Chicago, IL

Research Associate; 20 hrs/week — Jan. 2024 – Sept. 2024

- Designed electronic circuits and 3D printed wearable devices to use with electric muscle stimulation.
- Programmed microcontrollers (Seeeduino, Arduino Nano, ATmega64) in **C and C++** to control hardware for research.
- Designed and assembled **printed circuit boards** (PCBs) designed to generate electrical impulses used for muscle stimulation using **KiCad**.

Institute for Astronomy, University of Hawai'i at Mānoa — Honolulu, HI

Astrophysics REU Fellow; 40 hrs/week — May 2023 – Aug. 2023

- Conducted statistical analyses of 10,000+ characteristics of M-giant stars using data from TESS and Kepler in **Python3**.
- Cross-compared oscillation periods obtained from space-based telescopes to ground-based transient surveys.
- Presentation[1]: Chun, A., Saunders, N., Huber, D. (2023) Testing the Asteroseismic Detection Limits of Ground-Based Transient Survey using Kepler and TESS

Subaru Telescope, National Astronomical Observatory of Japan — Hilo, HI

Research Intern; 40 hrs/week — Jul. 2022 – Sept. 2022

- Characterized the AO3000, a new continuous surface deformable mirror that had 4,096 actuators.
- Created an automated data collection system controlling 1) movement of the deformable mirror, 2) data collection of laser interferometer, 3) data processing and saving in **Python3**, and 4) presented to parent company, ALPAO.
- Publication[2]: Lozi, L., Ahn, K., et al. including Chun, A., (2024). AO3k at Subaru: First on-sky results of the facility extreme-AO. arXiv:2407.19188

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[1] https://www.youtube.com/watch?v=hNwKU7kQk0Q

[2] https://arxiv.org/abs/2407.19188

UNIVERSITY PROJECTS

Mobile Theremin Mar. 2024

- Created a mobile theremin Apple application by using the True Depth sensor to control volume and the gyroscope rotation on the apple watch to control pitch using **Swift**

RhythEMS — Wearable 'Loop Pedal'[3] Mar. 2024

- Worked with a partner, Elena Hertel, to design and build a wearable device that incorporates EMS and a loop pedal.
- Developed the circuitry for the device, created the final protoboard, and wrote the code to control the EMS output.

Other projects include:

CPU Simulator (in C & Assembly) | TCP Implementation (in C) | IRC Chat Server (in C)| IP Router Simulator (in C)

LEADERSHIP EXPERIENCE

National Space Society, SpacEdge Chicago, IL

Metcalf Intern Sept. 2024 – Jan. 2025

- Research and develop courses for topics such as microgravity and planetary defense.

Le Vorris and Vox Circus Chicago, IL

President, Acrobatics Instructor March 2022 – May 2025

- Direct, perform, and write three circus shows, each with 20+ performers, every year for three years for 300+ people.
- Reserve spaces, budgeting finances, bridge communication between faculty, performers, and tech staff.

SKILLS & INTERESTS

Technical Skills: Git, Arduino IDE, AutoCAD, KiCad, Siemens NX, Sklearn (ML/AI), Powershell, Linux, Microsoft Office, Excel, Google Workspace, Protocol analyzer (WireShark), Docker

Laboratory Skills: Arduino, Printed Circuit Board Design, Oscilloscope, Optical Design, Digital Logic Analyzer, Multimeter, Arduino, Printed Circuit Board Design, Oscilloscope, Optical Design, Digital Logic Analyzer, Field Programmable Gate Arrays, Tapping & Dieing

Programming Skills: Java, C, C++, Python3, iOS/Swift, HTML/CSS, Assembly

Languages: English; Intermediate Proficiency in Spanish; Beginner Proficiency in German and Portuguese.

Interests: Tennis, Piano, Saxophone, Aerial Silks, Lyra, Partner Acrobatics, Electronic Music

[3] https://www.youtube.com/watch?v=A-OjZe4ul9s